1. Detection Performance Comparison

Given Data

Model	True Positives (TP)	False Positives (FP)	False Negatives (FN)
A – Viola-Jones	80	20	30
B – YOLO	90	15	20

(a) Precision and Recall

Formulas:

Precision = TP / (TP + FP)

Recall = TP / (TP + FN)

Model A – Viola-Jones

Precision:

= 80 / (80 + 20)

= 80 / 100

= **0.80 = 80%**

Recall:

= 80 / (80 + 30)

= 80 / 110

= **0.7273 = 72.73%**

Model B – YOLO

Precision:

= 90 / (90 + 15)

= 90 / 105

= **0.8571 = 85.71%**

Recall:

= 90 / (90 + 20)

= 90 / 110

= **0.8182 = 81.82%**

(b) Comparison

Metric	**Viola-Jones**	**YOLO**
Precision	80%	**85.71%**
Recall	72.73%	**81.82%**

Conclusion

YOLO performs better than Viola-Jones in both precision and recall. YOLO has 85.71% precision compared with 80% for Viola-Jones and 81.82% recall compared with 72.73%.

Therefore, YOLO is preferable for object detection because it produces more correct detections while also missing fewer objects.

4. Depth Estimation Comparison – Stereo vs SfM

Given Data

Object	Stereo Vision	Structure from Motion	Actual Depth
A	2.0 m	2.5 m	2.0 m
B	3.0 m	3.8 m	3.0 m
C	4.0 m	5.0 m	4.0 m

(a) Absolute Error

Formula:

Absolute Error = |Estimated Depth − Actual Depth|

Stereo Vision

Object A:

|2.0 − 2.0| = **0 m**

Object B:

|3.0 − 3.0| = **0 m**

Object C:

|4.0 − 4.0| = **0 m**

Average absolute error:

= (0 + 0 + 0) / 3

= **0 m**

Structure from Motion

Object A:

|2.5 − 2.0| = **0.5 m**

Object B:

|3.8 − 3.0| = **0.8 m**

Object C:

|5.0 − 4.0| = **1.0 m**

Average absolute error:

= (0.5 + 0.8 + 1.0) / 3

= **0.7667 m**

(b) Comparison

Method	**Average Absolute Error**
Stereo Vision	**0 m**
Structure from Motion	0.7667 m

Conclusion

Stereo Vision is more accurate because its average absolute error is 0 m, while Structure from Motion has an average error of 0.7667 m.

Therefore, Stereo Vision is preferable when depth accuracy is the main priority. Structure from Motion has a larger estimation error based on the given data.

6. Precision-Recall Trade-off Analysis

Given Data

Model	**TP**	**FP**	**FN**
A	70	30	20
B	85	25	25

(a) Precision and Recall

Formulas:

Precision = TP / (TP + FP)

Recall = TP / (TP + FN)

Model A

Precision:

= 70 / (70 + 30)

= 70 / 100

= **0.70 = 70%**

Recall:

= 70 / (70 + 20)

= 70 / 90

= **0.7778 = 77.78%**

Model B

Precision:

= 85 / (85 + 25)

= 85 / 110

= **0.7727 = 77.27%**

Recall:

= 85 / (85 + 25)

= 85 / 110

= **0.7727 = 77.27%**

(b) Comparison

Metric	**Model A**	**Model B**
Precision	70%	**77.27%**
Recall	**77.78%**	77.27%

Model A has slightly higher recall, while Model B has considerably higher precision.

Model B has nearly equal precision and recall, indicating a more balanced performance.

Conclusion

Model B is more balanced because its precision and recall are both 77.27%. Model A has a larger difference between precision and recall.

Therefore, Model B is preferable when both correct detection and detection completeness are important.

7. F1-Score Comparison

Given Data

Model	Precision	Recall
A	0.80	0.60
B	0.75	0.75

(a) F1-Score

Formula:

F1 = 2 × (Precision × Recall) / (Precision + Recall)

Model A

F1:

$= 2 \times (0.80 \times 0.60) / (0.80 + 0.60)$

$= 0.96 / 1.40$

= **0.6857**

Therefore:

F1-score = 68.57%

Model B

F1:

$= 2 \times (0.75 \times 0.75) / (0.75 + 0.75)$

$= 1.125 / 1.50$

= **0.75**

Therefore:

F1-score = 75%

(b) Comparison

Model	Precision	Recall	F1-Score
A	80%	60%	68.57%
B	75%	75%	**75%**

Conclusion

Model B has a higher F1-score of 75%, compared with 68.57% for Model A.

Although Model A has higher precision, its recall is considerably lower. Model B provides a better balance between precision and recall.

Therefore, Model B is the better overall model based on F1-score.

12. IoU Comparison

Given Data

Object	Model A	Model B
1	0.60	0.80
2	0.70	0.85
3	0.65	0.90

(a) Average IoU

Formula:

Average IoU = Sum of IoU values / Number of objects

Model A

Average IoU:

= (0.60 + 0.70 + 0.65) / 3

= 1.95 / 3

= **0.65**

Therefore:

Average IoU = 65%

Model B

Average IoU:

= (0.80 + 0.85 + 0.90) / 3

= 2.55 / 3

= **0.85**

Therefore:

Average IoU = 85%

(b) Comparison

Model	**Average IoU**
A	0.65 = 65%
B	**0.85 = 85%**

Conclusion

Model B has an average IoU of 85%, while Model A has only 65%. A higher IoU indicates better overlap between the predicted bounding box and the actual object region.

Therefore, Model B provides better detection and localization quality than Model A and should be preferred for applications where accurate object localization is important.